UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. RGB is an additive colour model.
2. Chroma key compositing replaces a colour background with another image.
3. Higher resolution always means a smaller file.
4. A scanner is an input device.
5. A web browser displays hypertext documents.
6. HTML is used to structure content on web pages.
7. Card-based authoring tools are best for messages with a clear beginning and end.
8. CMYK is used for printed output.
9. Multimedia combines text, graphics, audio, video and animation.
10. 3D animation does not use modelling.
11. Keyframes define important positions in an animation.
12. A bitmap image is made up of pixels.
13. A linear presentation allows users to choose their own path.
14. A prototype is an early model of a product.
15. Immersion is not important in virtual reality.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Card-based tool	B. Serif	C. Scanner
D. Chroma key	E. Frame rate	F. Copyright
G. Cel animation	H. Bitmap	I. HTML
J. Browser	K. Storyboard	L. Plotter
M. Multimedia	N. Tweening	O. Keyframe

1. software that displays web pages
2. important frame in an animation
3. input device that digitises documents
4. generating in-between frames
5. frames displayed per second
6. hand-drawn frame-by-frame animation
7. visual sequence plan
8. authoring tool organised like pages or cards
9. output device for large drawings
10. replacing a colour background
11. exclusive rights of a creator
12. font with small strokes at character ends
13. language for web structure
14. image made of pixels
15. integration of text, graphics, audio, video and animation

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Which is an input device?
   A. Plotter
   B. Monitor
   C. Scanner
   D. Printer

2. Amplitude of a sound wave relates to:
   A. Pitch
   B. Colour
   C. Loudness
   D. Frame rate

3. Target audience refers to:
   A. Colour model
   B. File format
   C. Intended users
   D. Software licence

4. Hertz measures:
   A. Loudness
   B. Resolution
   C. File size
   D. Frequency

5. A bitmap image is made of:
   A. Pixels
   B. Text characters
   C. Sound samples
   D. Mathematical objects

6. Which is an example of linear multimedia?
   A. A cinema presentation
   B. A website with menus
   C. An interactive kiosk
   D. A game with choices

7. Interlacing:
   A. Draws every line at once
   B. Adds sound
   C. Draws alternate lines of a frame
   D. Removes colour

8. RAM is:
   A. A web browser
   B. Permanent storage
   C. An input device
   D. Temporary working memory

9. Which is an audio file format?
   A. WAV
   B. JPEG
   C. HTML
   D. MPEG

10. Chroma key is used to:
    A. Compress audio
    B. Add text
    C. Scan images
    D. Replace a colour background

11. RGB is used mainly for:
    A. Video editing only
    B. Printed output
    C. Screen display
    D. Audio recording

12. Kerning adjusts:
    A. Line length
    B. Page size
    C. Font colour
    D. Space between individual character pairs

13. Which is an authoring tool category?
    A. Card-based
    B. JPEG
    C. RGB
    D. CMYK

14. A prototype is:
A. A final release
B. A font
C. An early model of a product
D. A codec

15. Hypermedia is best described as:
A. Text-only media
B. Multimedia with hyperlinks
C. A printed manual
D. A single video file

16. Which best defines multimedia?
A. Text, graphics, audio, video and animation
B. Text only
C. A printed book
D. A single image

17. Resolution refers to:
A. Number of pixels in an image
B. Colour name
C. File type
D. Font size

18. Sampling converts:
A. Images to vectors
B. Analog sound to digital values
C. Digital video to text
D. Text to speech

19. A serif font has:
A. No strokes
B. Only uppercase letters
C. Small strokes at character ends
D. No spacing

20. A higher sampling rate generally:
A. Changes colour
B. Improves accuracy
C. Removes audio
D. Reduces accuracy

21. Path animation moves an object:
A. Only vertically
B. Along a predefined route
C. Randomly
D. Without motion

22. Which is NOT a multimedia element?
A. Compiler
B. Audio
C. Text
D. Animation

23. MIDI stores:
A. Musical notes and instructions
B. Digitized audio samples
C. Video frames
D. Images

24. Streaming allows media to:
A. Play while downloading
B. Require full download first
C. Remove compression
D. Only play offline

25. Interactivity in multimedia means:
A. There is no navigation
B. Only text is shown
C. The user can control content and flow
D. The system plays automatically

26. A presentation where the user controls the sequence is called:
A. Printed
B. Non-linear
C. Static
D. Linear

27. Which is an output device?
A. Scanner
B. Plotter
C. Keyboard
D. Mouse

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28. A storyboard is:

A. A sound file
B. A video codec
C. A visual sequence plan
D. A colour model

29. Which is a benefit of multimedia in education?

A. It combines visuals and sound
B. It removes all interaction
C. It uses only text
D. It replaces all teachers

30. Keyframes are:

A. Web links
B. Colour profiles
C. Audio samples
D. Important frames in an animation

PART D (40 MARKS)

QUESTION 1

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 2

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 3

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(2 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(5 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

END OF QUESTION PAPER